



Sydney Girls High School

12 MAY

2010

TRIAL HIGHER SCHOOL
CERTIFICATE EXAMINATION

Mathematics Extension 2

This is a trial paper ONLY. It does not necessarily reflect the format or the content of the 2010 HSC Examination Paper in this subject.

General Instructions

- Reading Time - 5 minutes
- Working time - 3 hours
- Attempt ALL questions
- ALL questions are of equal value
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Standard integrals are supplied
- Board-approved calculators may be used.
- Diagrams are not to scale
- Each question attempted should be started on a new page. Write on one side of the paper only.

2 0 0 5 9 9 1 5

Candidate Number

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

Question One (15 marks)

Marks

a) Find $\int \frac{\sin x}{\cos^3 x} dx$. 2

b) Find $\int \frac{4x^3 - 2x^2 + 1}{2x - 1} dx$. 3

c) i) Find A , B and C given that $\frac{4x-6}{(x+1)(2x^2+3)} = \frac{A}{x+1} + \frac{Bx+C}{2x^2+3}$. 3

ii) Hence, find $\int \frac{4x-6}{(x+1)(2x^2+3)} dx$. 1

d) Find $\int \sin^{-1} x dx$. 3

e) Find $\int \frac{1}{3+2\cos\theta} d\theta$. 3

Question Two (15 marks)

Marks

a) If $z = \sqrt{3} - i$ and $w = 1 + i$, find:

i) zw 1

ii) $\arg z$ 1

iii) $|w'|$ 1

iv) $\operatorname{Im}\left(\frac{z}{w}\right)$ 2

b) $OPQR$ is a rectangle on the Argand diagram labelled anti-clockwise where O represents the origin and point P represents the complex number $3 + 4i$. Find the complex number representing Q and R given that $PQ = 2QR$. 2

c) i) Find the square roots of $21 + 20i$. 2

ii) Hence, solve $(1+i)z^2 + z - 5 = 0$. 3

d) The complex number z is such that $|z-1| = \operatorname{Re}(z)$.

i) Find the cartesian equation of the locus of z . 2

ii) Find the range of values of $|z|$. 1

Question Three (15 marks)

Marks

- a) The region bounded by $y = \log_e x$, $y = 1$ and $x = 3$ is rotated about the y axis.

i) Sketch this region on the number plane.

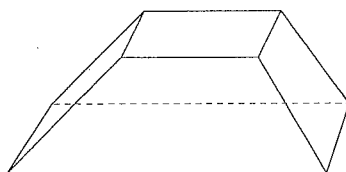
1

ii) Find the volume formed using the method of cylindrical shells.

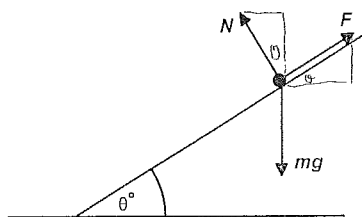
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- b) A solid is formed with the base and top both rectangles parallel to each other and 6 cm apart. The dimensions of the base are 11 cm and 15 cm and the dimensions of the top are 7 cm and 10 cm. If all other faces are trapeziums, find the volume of the solid.

5



- c) An object of mass m is lying on an inclined plane at an angle θ to the horizontal. As shown in the diagram below, the object is subject to a gravitational force mg , a normal reaction force N and a frictional force F .



The object is not moving.

Resolve the forces acting on the object, and hence find an expression for

$\frac{F}{N}$ in terms of θ .

3

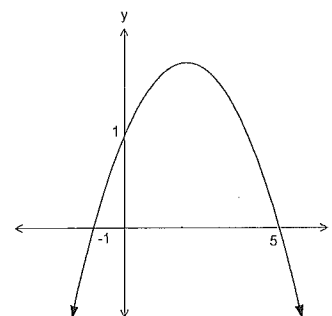
- d) Find $\frac{dy}{dx}$ given $x^3 + x^2y^4 = 0$.

3

Question Four (15 marks)

Marks

- a) The diagram shows the graph of $y = f(x)$.



Draw separate one-third page sketches of the graphs of the following :

i) $y = |f(x)|$

1

ii) $y = f(|x|)$

1

iii) $y = [f(x)]^2$

2

iv) $y = e^{f(x)}$

2

v) $y = -\frac{1}{f(x)}$

3

- b) The equation $x^3 + 3x^2 - 2x - 2 = 0$ has roots α , β and γ .

Find the equation with roots $\frac{2\alpha}{\beta\gamma}$, $\frac{2\beta}{\alpha\gamma}$ and $\frac{2\gamma}{\alpha\beta}$.

3

- c) Determine the greatest and least values of $\arg(z)$ if $|z - 4i| = 2$.

3

Question Five (15 marks)

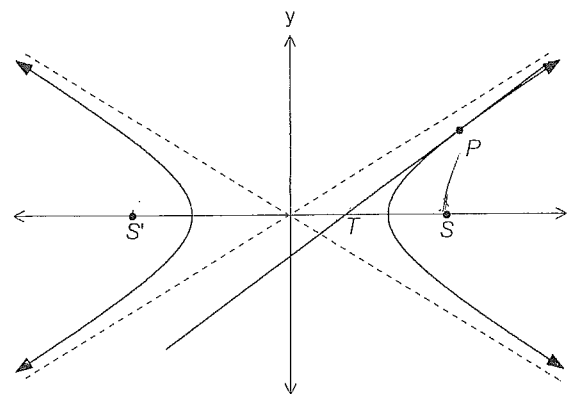
Marks

- a) Given $1-i$ is a root of $x^3 - 3x^2 + 4x - 2 = 0$ find the other roots. 2
- b) For the equation $x^4 + x^3 - 16x^2 - 4x + 48 = 0$, the product of two of the roots is 6.
- i) Hence, express the equation in the form $(x^2 + ax + b)(x^2 + cx + d) = 0$. 3
- ii) Find the roots of the equation. 1
- c) i) Given $x = \alpha$ is a double root of the equation $ax^4 + 4bx + c = 0$, deduce that $\alpha^3 = -\frac{b}{a}$. 2
- ii) Also, deduce that $ac^3 = 27b^4$. 3
- iii) Hence or otherwise, solve the equation $27x^4 - 32x + 16 = 0$, given that it has a double root. 4

Question Six (15 marks)

Marks

- a) $P\left(p, \frac{1}{p}\right)$ and $Q\left(q, \frac{1}{q}\right)$ are two points on the rectangular hyperbola $xy = 1$.
- i) Derive the equation of the chord PQ and show that it can be expressed in general form as $x + pqy - (p + q) = 0$. 2
- ii) Hence, show that the area of $\triangle OPQ$ is $\frac{|p^2 - q^2|}{2|pq|}$ units². 4
- b) The point $P(x_1, y_1)$ lies on the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$. The tangent at P cuts the x -axis at T .

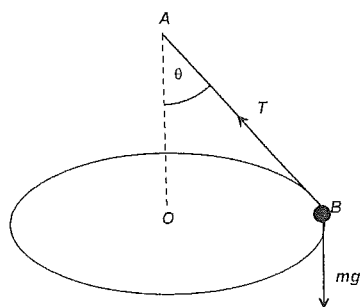


- i) Find the coordinates of the foci S and S' . 1
- ii) Show that the equation of the tangent at P is given by $\frac{xx_1}{16} - \frac{yy_1}{9} = 1$. 3
- iii) Show that $\frac{S'T}{ST} = \frac{S'P}{SP}$. 3
- iv) Hence, deduce that $\angle S'PT = \angle SPT$. 2

Question Seven (15 marks)

Marks

- a) A particle of mass m kg is attached to one end of a light string at B . The other end of the string is fixed at a point A . The particle rotates in a horizontal circle of radius r metres at g rad/s, the centre of the circle being directly below A .



The forces acting on the particle are the tension in the string T and the gravitational force mg .

Let $\angle BAO = \theta$.

- i) Show that $T \sin \theta = mg^2 r$. 1
- ii) Prove that $\theta = \tan^{-1}(gr)$. 1
- iii) Prove that $T = mg\sqrt{1 + g^2 r^2}$. 2
- b) i) Use De Moivre's Theorem to express $\cos 4\theta$ and $\sin 4\theta$ as powers of $\cos \theta$ and $\sin \theta$. 2
- ii) Hence show that $\tan 4\theta = \frac{4t - 4t^3}{1 - 6t^2 + t^4}$ where $t = \tan \theta$. 1
- iii) By first solving the equation $\tan 4\theta = 1$ for $0 \leq \theta \leq 2\pi$, solve the equation $x^4 + 4x^3 - 6x^2 - 4x + 1 = 0$. 3
- iv) Hence find the value of $\tan \frac{\pi}{16} \tan \frac{3\pi}{16} \tan \frac{5\pi}{16} \tan \frac{7\pi}{16}$. 2
- c) Evaluate $\int_0^{\pi} x \cos 2x \, dx$. 3

Question Eight (15 marks)

Marks

- a) i) Use integration by parts to show that $I_n = \frac{n-1}{n} I_{n-2}$ for $n \geq 2$ given $I_n = \int_0^{\frac{\pi}{2}} (\sin x)^n \, dx$ where n is an integer ($n \geq 0$). 3
- ii) Deduce $I_{2n} = \frac{2n-1}{2n} \times \frac{2n-3}{2n-2} \times \dots \times \frac{3}{4} \times \frac{1}{2} \times \frac{\pi}{2}$. 3
- b) Given the identity $\cos(A+B) + \cos(A-B) = 2 \cos A \cos B$, solve the equation $\cos 5x + \cos 3x + 2 \cos x = 0$ for $0 \leq x \leq \frac{\pi}{2}$. 3
- c) Two sides of a triangle are of length $2x$ cm and $3x$ cm. The angles opposite these sides differ by 45° . Show that the smaller of the two angles is given by $\tan^{-1}\left(\frac{2+3\sqrt{2}}{7}\right)$. 4
- d) The positive integers are bracketed as follows $(1), (2, 3), (4, 5, 6), (7, 8, 9, 10), \dots$. The n th bracket has n integers. Prove that the sum of the integers in the n th bracket is $\frac{n}{2}(n^2 + 1)$. 2

End of paper

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QUESTION 1

a. Let $u = \cos x$

$$\frac{du}{dx} = -\sin x$$

$$\begin{aligned}\int \frac{\sin x}{\cos^3 x} dx &= \int -u^{-3} du \\ &= -\frac{u^{-2}}{-2} + C \\ &= \frac{1}{2\cos^2 x} + C\end{aligned}$$

b.

$$\frac{2x^2}{2x-1} \sqrt{4x^3-2x^2+1}$$

$$\frac{4x^3-2x^2}{4x^3-2x^2} \cdot \frac{2x^2}{0+1}$$

$$\begin{aligned}\int \frac{4x^3-2x^2+1}{2x-1} dx &= \int \left(2x^2 + \frac{1}{2x-1} \right) dx \\ &= \frac{2x^3}{3} + \frac{1}{2} \ln(2x-1) + C\end{aligned}$$

c.

$$i. 4x-6 = A(2x^2+3) + (Bx+C)(x+1)$$

$$\text{let } x = -1$$

$$-10 = 5A$$

$$\therefore A = -2$$

$$0 = 2A + B$$

$$\therefore B = 4$$

$$-6 = 3A + C$$

$$\therefore C = 0$$

ii.

$$\begin{aligned}\int \frac{4x-6}{(x+1)(2x^2+3)} dx &= \int \frac{-2}{x+1} + \frac{4x}{2x^2+3} \\ &= -2\ln(x+1) + \ln(2x^2+3) + C\end{aligned}$$

d.

let

$$u = \sin^{-1} x \quad dv = dx$$

$$u' = \frac{1}{\sqrt{1-x^2}} \quad v = x$$

$$\int \sin^{-1} x dx = x \sin^{-1} x - \int \frac{x}{\sqrt{1-x^2}} dx$$

Let

$$u = 1-x^2$$

$$du = -2x dx$$

$$-\frac{1}{2} du = x dx$$

$$\begin{aligned}x \sin^{-1} x - \int \frac{x}{\sqrt{1-x^2}} dx &= x \sin^{-1} x - \int \frac{-du}{2\sqrt{u}} \\ &= x \sin^{-1} x + \frac{1}{2} \int u^{-\frac{1}{2}} \\ &= x \sin^{-1} x + \frac{1}{2} \times 2u^{\frac{1}{2}} \\ &= x \sin^{-1} x + u^{\frac{1}{2}} \\ &= x \sin^{-1} x + \sqrt{1-x^2}\end{aligned}$$

e.

$$\int \frac{1}{3+2\cos\theta} d\theta = \int \frac{1}{3+2\left(\frac{1-t^2}{1+t^2}\right)} \cdot \frac{2dt}{1+t^2}$$

$$= \int \frac{1}{\frac{5+t^2}{1+t^2}} \cdot \frac{2dt}{1+t^2}$$

$$= \int \frac{2}{5+t^2} dt$$

$$= 2 \left(\frac{1}{\sqrt{5}} \tan^{-1} \frac{t}{\sqrt{5}} \right) + C$$

$$= 2 \left(\frac{1}{\sqrt{5}} \tan^{-1} \frac{\tan \frac{\theta}{2}}{\sqrt{5}} \right) + C$$

QUESTION 2

a.

i.

$$\begin{aligned}zw &= (\sqrt{3}-i)(1+i) \\ &= \sqrt{3}-i^2+i(\sqrt{3}-1) \\ &= \sqrt{3}+1+i(\sqrt{3}-1)\end{aligned}$$

ii.

$$\begin{aligned}\arg(z) &= -\tan^{-1}\left(\frac{1}{\sqrt{3}}\right) \\ &= -\frac{\pi}{6}\end{aligned}$$

iii.

$$\begin{aligned}|w| &= \sqrt{1^2+1^2} \\ &= \sqrt{2} \\ |w^7| &= |w|^7 \\ &= (\sqrt{2})^7 \\ &= 8\sqrt{2}\end{aligned}$$

iv.

$$\begin{aligned}\operatorname{Im}\left(\frac{z}{w}\right) &= \operatorname{Im}\left(\frac{\sqrt{3}-i}{1+i} \times \frac{1-i}{1-i}\right) \\ &= -\frac{\sqrt{3}+1}{2}\end{aligned}$$

b.

$$R = 2i(3+4i)$$

$$= -8+6i$$

$$Q = \overline{OR} + \overline{OP}$$

$$= -8+6i+3+4i$$

$$= -5+10i$$

c.

i.

$$\begin{aligned}\sqrt{21+20i} &= a+ib \\ 21+20i &= a^2+2abi-b^2 \\ a^2-b^2 &= 21 \\ 2ab &= 20 \\ a &= \pm 5 \\ b &= \pm 2 \\ \therefore \pm(5+2i)\end{aligned}$$

ii.

$$\begin{aligned}z &= \frac{-1 \pm \sqrt{1^2+20(1+i)}}{2(1+i)} \\ &= \frac{-1 \pm (5+2i)}{2(1+i)} \\ &= \frac{4+2i}{2(1+i)} \quad \text{or} \quad \frac{-6-2i}{2(1+i)} \\ &= \frac{2+i}{1+i} \quad \text{or} \quad \frac{-3-i}{1+i}\end{aligned}$$

d.

i.

$$|z-1| = \operatorname{Re}(z)$$

$$\text{If } z = x+iy$$

$$|(x-1)+iy| = x$$

$$\sqrt{(x-1)^2+y^2} = x$$

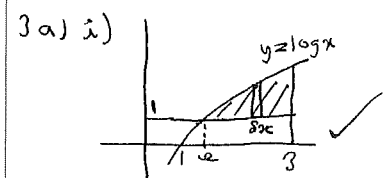
$$(x-1)^2+y^2 = x^2$$

$$x^2-2x+1+y^2 = x^2$$

$$y^2 = 2x-1 \quad \text{or} \quad x = \frac{1}{2}(y^2+1)$$

ii.

$$|z| \geq \frac{1}{2}$$



$$\begin{aligned} \text{ii) } V_{\text{shell}} &= \pi(R^2 - r^2) \\ &= \pi((x + \delta x)^2 - x^2)(y - 1) \\ &= 2\pi x(y - 1)\delta x \end{aligned}$$

$$\begin{aligned} V_{\text{solid}} &= \lim_{\delta x \rightarrow 0} \sum_{x=2}^3 2\pi x(y - 1)\delta x \\ &= 2\pi \int_2^3 x(y - 1) dx \\ &= 2\pi \int_2^3 (x \log x - x) dx \\ &= 2\pi \left[\frac{1}{2} x^2 \log x - \frac{1}{4} x^2 \right]_2^3 \end{aligned}$$

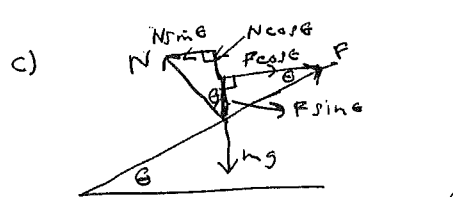
$$\begin{aligned} \text{Let } u &= \log x \quad v' = x \\ u' &= \frac{1}{x} \quad v = \frac{x^2}{2} \end{aligned}$$

$$\begin{aligned} V_{\text{solid}} &= 2\pi \left(\left[\log x \cdot \frac{x^2}{2} \right]_2^3 - \frac{1}{2} \int_2^3 x^2 \cdot \frac{1}{x} dx - \left[\frac{x^2}{2} \right]_2^3 \right) \\ &= 2\pi \left(\frac{9 \log 3}{2} - \frac{3}{2} \left[\frac{x^2}{2} \right]_2^3 - \frac{x^2}{2} \right) \\ &= 2\pi \left(\frac{9 \log 3}{2} - \frac{27}{4} + \frac{2}{4} \right) \\ &= \pi \left(9 \log 3 - \frac{25}{2} \right) \text{ cm}^3 \end{aligned}$$

$$\begin{aligned} \text{b) } V_{\text{shell}} &= A \delta h \\ &= xy \delta h \end{aligned}$$

$$\begin{aligned} V_{\text{solid}} &= \lim_{\delta h \rightarrow 0} \sum_{h=0}^6 xy \delta h \\ &= \int_0^6 xy dh \\ &= \int_0^6 \left(\frac{2h}{3} + 7 \right) \left(\frac{5h}{6} + 10 \right) dh \\ &= \int_0^6 \left(\frac{5h^2}{9} + \frac{75h}{6} + 70h \right) dh \\ &= \left[\frac{5h^3}{27} + \frac{25h^2}{4} + 70h \right]_0^6 \\ &= \frac{5 \times 6^3}{27} + \frac{25 \times 6^2}{4} + 70 \times 6 \\ &= 685 \text{ cm}^3 \end{aligned}$$

$h=0 \quad x=7 \quad y=10$
 $h=6 \quad x=11 \quad y=15$
 $x = mh + b \quad y = mh + b$
 $7 = b \quad 10 = b$
 $11 = 6m + 7 \quad 15 = 6m + 10$
 $4 = 6m \quad 5 = 6m$
 $m = \frac{2}{3} \quad m = \frac{5}{6}$



$$\begin{aligned} F \sin \theta + N \cos \theta - mg &= 0 \\ N \sin \theta - F \cos \theta &= 0 \\ F \sin^2 \theta + N \sin \theta \cos \theta &= mg \sin \theta \\ N \sin \theta \cos \theta - F \cos^2 \theta &= 0 \end{aligned}$$

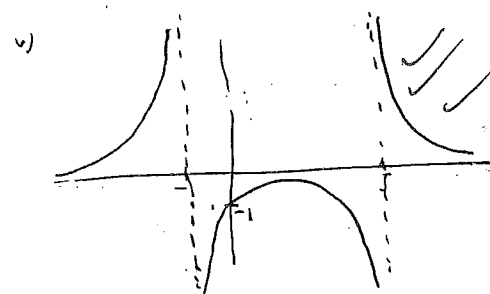
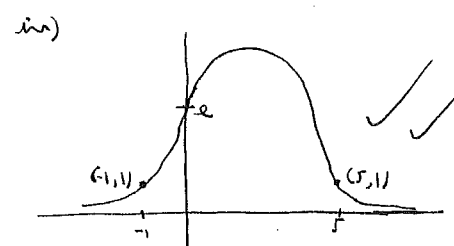
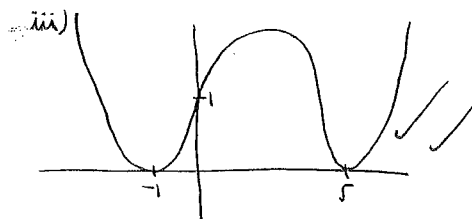
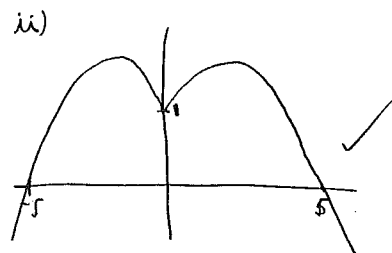
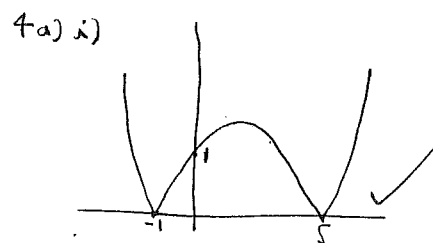
$$\therefore F = mg \sin \theta$$

$$\begin{aligned} F \sin \theta \cos \theta + N \cos^2 \theta &= mg \cos \theta \\ N \sin^2 \theta - F \sin \theta \cos \theta &= 0 \end{aligned}$$

$$\therefore N = mg \cos \theta$$

$$\begin{aligned} \frac{F}{N} &= \frac{mg \sin \theta}{mg \cos \theta} \\ &= \tan \theta \end{aligned}$$

$$\begin{aligned} \text{d) } 3x^2 + x^2 \cdot 4y^3 \frac{dy}{dx} + 2xy^4 &= 0 \\ \frac{dy}{dx} &= \frac{-2xy^4 - 3x^2}{4x^2y^3} \\ &= -\frac{2y^4 + 3x}{4xy^3} \end{aligned}$$



$$\begin{aligned} \text{(A) } \frac{2x^2}{x^2y} &= \frac{2x}{y} \quad \angle \beta \gamma = -\frac{-2}{1} \\ &= 2 \end{aligned}$$

$$\therefore \text{not are } x^2, y^2 \quad \checkmark$$

$$(\sqrt{x})^3 + 3(\sqrt{x})^2 - 2\sqrt{x} - 2 \geq 0$$

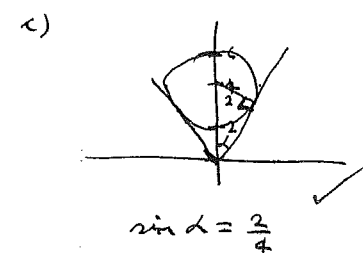
$$x\sqrt{x} + 3x - 2\sqrt{x} - 2 \geq 0 \quad \checkmark$$

$$x\sqrt{x} - 2\sqrt{x} = 2 - 3x$$

$$x(x-2)^2 = (2-3x)^2$$

$$x^3 - 4x^2 + 4x = 4 - 12x + 9x^2$$

$$x^3 - 13x^2 + 16x - 4 \geq 0 \quad \checkmark$$



$$\sin \alpha = \frac{2}{4}$$

$$\therefore \alpha = \frac{\pi}{6}$$

$$\begin{aligned} \text{greatest arg } z &= \frac{\pi}{2} + \frac{\pi}{6} \\ &= \frac{2\pi}{3} \quad \checkmark \end{aligned}$$

$$\begin{aligned} \text{least arg } z &= \frac{\pi}{2} - \frac{\pi}{6} \\ &= \frac{\pi}{3} \quad \checkmark \end{aligned}$$

Question Five

(a) If $1-i$, then $1+i$ must also be a root.

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha + 1 - i + 1 + i = -\frac{(-3)}{1}$$

$$\alpha + 3 - 2 = 1$$

Hence, roots are $1-i, 1+i, 1$.

(b)(i) $\alpha\beta = 6$

$$\alpha\beta\gamma\delta = 48 \Rightarrow \gamma\delta = 8$$

$$(x^2 + ax + 6)(x^2 + cx + 8) = 0$$

$$\text{coefficient of } x^3 \quad a + c = 1$$

$$\text{coefficient of } x \quad 8a + 6c = -4$$

$$8a + 8c = 8$$

$$2a = -10 \Rightarrow a = -5, c = 6$$

$$(x^2 - 5x + 6)(x^2 + 6x + 8) = 0$$

(b)(ii) $(x-3)(x-2)(x+4)(x+2) = 0$

$$\therefore x = 3, 2, -4 \text{ or } -2$$

(c)(i) $P(x) = ax^4 + 4bx + c$

$$P'(x) = 4ax^3 + 4b$$

double root at $x = \alpha$

$$\Rightarrow P(\alpha) = P'(\alpha) = 0$$

$$4a\alpha^3 + 4b = 0$$

$$4a\alpha^3 = -4b$$

$$\alpha^3 = \frac{-4b}{4a} = -\frac{b}{a}$$

(c)(ii) $P(\alpha) = 0$

$$a\alpha^4 + 4b\alpha + c = 0$$

$$\alpha(a\alpha^3 + 4b) = -c$$

$$\alpha\left(a \times -\frac{b}{a} + 4b\right) = -c$$

$$\alpha(-b + 4b) = -c$$

$$\alpha^3(3b) = -c^3$$

$$-\frac{b}{a} \times 27b^3 = -c^3$$

$$\therefore ac^3 = 27b^4$$

(c)(iii) $a = 27, b = -8, c = 16$

$$\alpha^3 = \frac{8}{27} \quad \therefore \alpha = \frac{2}{3}$$

$$27x^4 - 32x + 16 = (3x-2)^2(3x^2 + 4x + 4)$$

$$x = \frac{-4 \pm \sqrt{16 - 4(12)}}{6} = \frac{-4 \pm \sqrt{-32}}{6}$$

$$\therefore x = \frac{2}{3}, \frac{2}{3}, \frac{-2 \pm 2\sqrt{2}i}{3}$$

Question Six

(a)(i) $m_{PQ} = \frac{\frac{1}{p} - \frac{1}{q}}{\frac{p}{p-q} - \frac{q}{pq(p-q)}} = -\frac{1}{pq}$

$$y - \frac{1}{p} = -\frac{1}{pq}(x - p)$$

$$pqy - q = -x + p$$

$$x + pqy - (p + q) = 0$$

(a)(ii) height (distance from O to PQ)

$$h = \frac{|0 + (pq)0 - (p + q)|}{\sqrt{1^2 + (pq)^2}}$$

$$= \frac{|p + q|}{\sqrt{1 + p^2q^2}}$$

$$PQ = \sqrt{(p-q)^2 + \left(\frac{1}{p} - \frac{1}{q}\right)^2}$$

$$= \sqrt{(p-q)^2 + \frac{1}{(pq)^2}(q-p)^2}$$

$$= \sqrt{(p-q)^2 \left(1 + \frac{1}{p^2q^2}\right)} \text{ as } (q-p)^2 = (p-q)^2$$

$$= \sqrt{\frac{(p-q)^2}{p^2q^2} \times \sqrt{p^2q^2 + 1}}$$

$$= \left|\frac{p-q}{pq}\right| \times \sqrt{1 + p^2q^2}$$

$$\text{Area} = \frac{1}{2} \times \frac{|p+q|}{\sqrt{1+p^2q^2}} \times \left|\frac{p-q}{pq}\right| \times \sqrt{1+p^2q^2}$$

$$= \frac{|p^2 - q^2|}{2|pq|} \text{ units}^2$$

(b)(i) $e^2 = 1 + \frac{9}{16} \quad \therefore e = \frac{5}{4}$

$$\text{Focus} = (\pm ae, 0) = (\pm 5, 0)$$

(b)(ii) differentiate wrt x

$$\frac{2x}{16} - \frac{2y}{9} \times \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{x}{8} \times \frac{9}{2y} = \frac{9x}{16y}$$

(b)(ii). continued at $P \quad m_T = \frac{9x_1}{16y_1}$

equation of tangent :

$$y - y_1 = \frac{9x_1}{16y_1}(x - x_1)$$

$$16y_1(y - y_1) = 9x_1(x - x_1)$$

$$16yy_1 - 16(y_1)^2 = 9xx_1 - 9(x_1)^2$$

$$\frac{xx_1}{16} - \frac{yy_1}{9} = \frac{(x_1)^2}{16} - \frac{(y_1)^2}{9}$$

i.e. $\frac{xx_1}{16} - \frac{yy_1}{9} = 1$ as (x_1, y_1) lies on $\frac{x^2}{16} - \frac{y^2}{9} = 1$

(b)(iii) T is x -int. where $y = 0$

$$T = \left(\frac{16}{x_1}, 0\right)$$

$$\frac{S'T}{ST} = \frac{5 + \frac{16}{x_1}}{5 - \frac{16}{x_1}}$$

i.e. $\frac{S'T}{ST} = \frac{5x_1 + 16}{5x_1 - 16}$

$$\frac{S'P}{SP} = \frac{ePM'}{ePM} \text{ where } M \text{ is corr. directrix}$$

$$= \frac{PM'}{PM} = \frac{x_1 + \frac{16}{5}}{x_1 - \frac{16}{5}}$$

i.e. $\frac{S'P}{SP} = \frac{5x_1 + 16}{5x_1 - 16} = \frac{S'T}{ST}$

(b)(iv) Let $\angle S'PT = \beta, \angle SPT = \gamma$ and $\angle PTS = \alpha$

$$\angle PTS' = 180 - \alpha \text{ (st. } \angle)$$

$$\frac{\sin(180 - \alpha)}{S'P} = \frac{\sin \beta}{S'T}$$

i.e. $\sin \beta = \frac{S'T \sin(180 - \alpha)}{S'P}$

$$= \frac{S'T \sin \alpha}{S'P}$$

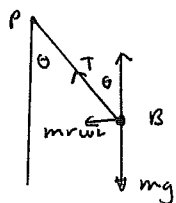
$$\text{Similarly, } \sin \gamma = \frac{ST \sin \alpha}{SP}$$

Since $\frac{S'T}{ST} = \frac{S'P}{SP}$ then $\frac{S'T}{S'P} = \frac{ST}{SP}$

Hence $\sin \beta = \sin \gamma$ i.e. $\beta = \gamma$



a)



1) $w = g$

Resolving Horizontally

$$T \sin \theta = m r \omega^2$$

$$T \sin \theta = m r g^{\frac{1}{2}} \quad (1)$$

ii) Balancing Vertically

$$T \cos \theta = m g \quad (2)$$

$$(1) \div (2) \quad \frac{T \sin \theta}{T \cos \theta} = \frac{m r g^{\frac{1}{2}}}{m g}$$

$$\tan \theta = g r$$

$$\therefore \theta = \tan^{-1}(g r)^{\frac{1}{2}} \quad (1)$$

iii) From (1) $T^2 \sin^2 \theta = m^2 r^2 g^{\frac{1}{2}} \quad (3)$

from (2) $T^2 \cos^2 \theta = m^2 g^2 \quad (4)$

$$(3) + (4) \quad T^2 (\sin^2 \theta + \cos^2 \theta) = m^2 g^2 + m^2 r^2 g^{\frac{1}{2}}$$

$$\begin{aligned} T^2 &= m^2 g^2 (1 + g^{\frac{1}{2}} r^2) \\ T &= m g \sqrt{1 + g^{\frac{1}{2}} r^2} \quad (2) \end{aligned}$$

b) i) $\cos 4\theta + i \sin 4\theta = (\cos \theta + i \sin \theta)^4$ [De Moivre Th]

$$\text{RHS} = \cos^4 \theta + 4i \cos^3 \theta \sin \theta - 6 \cos^2 \theta \sin^2 \theta - 4i \cos \theta \sin^3 \theta + \sin^4 \theta$$

$$\text{Equate Re } \cos 4\theta = \cos^4 \theta - 6 \cos^2 \theta \sin^2 \theta + \sin^4 \theta \quad (1)$$

$$\text{Equate Im } \sin 4\theta = 4 \cos^3 \theta \sin \theta - 4 \cos \theta \sin^3 \theta \quad (2)$$

$$\begin{aligned} \text{ii) } (2) \div (1) \quad \tan 4\theta &= \frac{4 \cos^3 \theta \sin \theta - 4 \cos \theta \sin^3 \theta}{\cos^4 \theta - 6 \cos^2 \theta \sin^2 \theta + \sin^4 \theta} \\ &= \frac{4x - 4x^3}{1 - 6x^2 + x^4} \quad \text{(Dividing top & bottom by } \cos^4 \theta) \end{aligned}$$

(1)

(1)



7b)

iii) $\tan 4\theta = 1$

$$4\theta = \frac{\pi}{4}, \frac{5\pi}{4}, \frac{9\pi}{4}, \frac{13\pi}{4} \quad \checkmark$$

$$\theta = \frac{\pi}{16}, \frac{5\pi}{16}, \frac{9\pi}{16}, \frac{13\pi}{16} \quad \checkmark$$

roots are $\tan \frac{\pi}{16}, \tan \frac{5\pi}{16}$

$$\tan \frac{9\pi}{16} = -\tan \frac{7\pi}{16}$$

$$\tan \frac{13\pi}{16} = -\tan \frac{3\pi}{16}$$

ie $\tan \frac{\pi}{16}, \tan \frac{5\pi}{16}, -\tan \frac{7\pi}{16}, -\tan \frac{3\pi}{16}$ or equivalent

iv) Product of roots = 1

$$\therefore \tan \frac{\pi}{16} \times \tan \frac{5\pi}{16} \times (-\tan \frac{7\pi}{16}) \times (-\tan \frac{3\pi}{16}) = 1$$

$$\text{or } \tan \frac{\pi}{16} \tan \frac{3\pi}{16} \tan \frac{5\pi}{16} \tan \frac{7\pi}{16} = 1 \quad (2)$$

(Note Hence find, A must follow from iii)

c) $I = \int_0^{\pi} x \cos 2x \, dx$ put $\pi - x$ in place of x

$$I = \int_0^{\pi} (\pi - x) \cos (2\pi - 2x) \, dx$$

$$I = \int_0^{\pi} \pi \cos (2\pi - 2x) \, dx - \int_0^{\pi} x \cos (2\pi - 2x) \, dx$$

$$I = \int_0^{\pi} \pi \cos (2x) \, dx - \int_0^{\pi} x \cos (2x) \, dx \quad \checkmark$$

$$= \pi \int_0^{\pi} \cos 2x \, dx - I$$

$$2I = -\pi [\sin 2x]_0^{\pi}$$

$$= -\pi [0 - 0]$$

$$= 0$$

$$\therefore I = 0$$

let $u = x$ $u' = \cos 2x$

$$u = 1 \quad u' = \frac{1}{2} \sin 2x$$

$$I = \left[\frac{x}{2} \sin 2x \right]_0^{\pi} - \int_0^{\pi} \frac{1}{2} \sin 2x \, dx$$

$$= 0 - \left[\frac{1}{4} \cos 2x \right]_0^{\pi}$$

$$= 0 + \frac{1}{4} [1 - 1]$$

$$= 0$$

Question Eight

$$a) 1) \quad I_n = \int_0^{\frac{\pi}{2}} (\sin x)^n dx$$

$$= \int_0^{\frac{\pi}{2}} (\sin x) (\sin x)^{n-1} dx$$

$$\text{let } u = \sin^{n-1} x \quad v = \sin x$$

$$du = (n-1)(\sin x)^{n-2} \cos x \quad dv = \cos x$$

$$I_n = -[\cos x \sin^{n-1} x]_0^{\frac{\pi}{2}} + (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2} x \cos^2 x dx$$

$$= 0 + (n-1) \int_0^{\frac{\pi}{2}} (\sin^{n-2} x) (1 - \sin^2 x) dx$$

$$= (n-1) \left[\int_0^{\frac{\pi}{2}} \sin^{n-2} x dx - \int_0^{\frac{\pi}{2}} \sin^n x dx \right]$$

$$= (n-1) [I_{n-2} - I_n]$$

$$I_n = (n-1)I_{n-2} - (n-1)I_n \quad (3)$$

$$n I_n = (n-1) I_{n-2}$$

$$I_n = \frac{n-1}{n} I_{n-2}$$

ii) put $2n$ in place of n

$$I_{2n} = \frac{2n-1}{2n} I_{2n-2}$$

$$= \frac{2n-1}{2n} \times \frac{2n-3}{2n-2} \times I_{2n-4} \dots$$

$$= \frac{2n-1}{2n} \times \frac{2n-3}{2n-2} \times \frac{2n-5}{2n-4} \times \dots \times I_2 \times I_0 \quad (3)$$

$$\text{and } I_0 = \int_0^{\frac{\pi}{2}} 1 dx = \frac{\pi}{2}, \quad I_2 = \int_0^{\frac{\pi}{2}} \sin^2 x dx$$

$$= \frac{1}{2} [1 - \cos 2x]_0^{\frac{\pi}{2}}$$

$$= \frac{1}{2} [(1-1) - (1-0)]$$

$$= \frac{1}{2}$$

$$\therefore I_{2n} = \frac{2n-1}{2n} \times \frac{2n-3}{2n-2} \times \frac{2n-5}{2n-4} \times \dots \times \frac{1}{2} \times \frac{\pi}{2}$$

Now numerator & denominator increase by 2 from right to left

$$\therefore I_{2n} = \frac{2n-1}{2n} \times \frac{2n-3}{2n-2} \times \frac{2n-5}{2n-4} \times \dots \times \frac{5}{6} \times \frac{3}{4} \times \frac{1}{2} \times \frac{\pi}{2}$$

$$Q 8 b) \quad \cos(A+B) + \cos(A-B) = 2 \cos A \cos B$$

$$\cos(4x+2) + \cos(4x-2) = 2 \cos 4x \cos 2$$

$\therefore$ eqn becomes

$$2 \cos 4x \cos 2 + 2 \cos 2 = 0 \quad \checkmark$$

$$2 \cos 2 (\cos 4x + 1) = 0$$

$$2 \cos 2 = 0, \quad \cos 4x + 1 = 0$$

$$\cos 2 = 0, \quad \cos 4x = -1$$

$$2 = \frac{\pi}{2}$$

$$4x = \pi$$

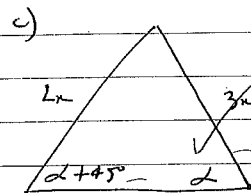
$$x = \frac{\pi}{4} \quad \checkmark$$

$$0 \leq x \leq \frac{\pi}{2}$$

$$0 \leq 4x \leq 2\pi$$



(3)



$$\frac{2x}{\sin \alpha} = \frac{3x}{\sin (\alpha + 45^\circ)} \quad \checkmark$$

$$\frac{\sin (\alpha + 45^\circ)}{\sin \alpha} = \frac{3}{2}$$

$$3 \sin \alpha = 2 \sin (\alpha + 45^\circ)$$

$$3 \sin \alpha = 2 (\sin \alpha \cos 45^\circ + \cos \alpha \sin 45^\circ)$$

$$3 \sin \alpha = 2 \left(\frac{\sin \alpha}{\sqrt{2}} + \frac{\cos \alpha}{\sqrt{2}} \right)$$

$$3 \sin \alpha = \sqrt{2} \sin \alpha + \sqrt{2} \cos \alpha$$

$$\sin \alpha (3 - \sqrt{2}) = \sqrt{2} \cos \alpha \quad \checkmark$$

$$\frac{\sin \alpha}{\cos \alpha} = \frac{\sqrt{2}}{3 - \sqrt{2}} \times \frac{3 + \sqrt{2}}{3 + \sqrt{2}}$$

(4)

$$\tan \alpha = \frac{3\sqrt{2} + 2}{7}$$

$$\alpha = \tan^{-1} \left(\frac{2 + 3\sqrt{2}}{7} \right) \quad \checkmark$$

↑ work for

same work

or diagram

8 d) (1), (2, 3), (4, 5, 6), (7, 8, 9, 10)

T_1 in first brackets = 1

T_1 in 2nd " = 1 + 1

T_1 in 3rd " = 1 + 1 + 2

T_1 in 4th " = 1 + 1 + 2 + 3

T_1 in 5th " = 1 + 1 + 2 + 3 + 4

∴ T_1 in n th " = 1 + (1 + 2 + 3 + ... + $n-1$)

$$\text{ie } = 1 + \frac{n-1}{2} (1 + n-1) \quad S_n = \frac{n}{2} (a + l)$$

$$= 1 + \frac{n^2 - n}{2}$$

$$\text{∴ } a = \frac{n^2 - n + 2}{2} \quad \checkmark$$

(2)

$$S_n = \frac{n}{2} (2a + (n-1)d)$$

$$\text{Sum of } n \text{ terms} = \frac{n}{2} \left(2 \left(\frac{n^2 - n + 2}{2} \right) + (n-1)1 \right) \checkmark$$

$$= \frac{n}{2} (n^2 - n + 2 + n - 1)$$

$$= \frac{n}{2} (n^2 + 1)$$